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AVAREBIOTECH

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Field Case Study Report of "Avare Biotech" app powered by MAKSA technology on Fresh Equine Semen (Texas, USA, 2025)

Date: November 2025
Location: Texas, USA

Conducted by:	Assisted by:	Verified by:
Gregg Veneklasen (Texas, USA)	Kylee Brown (Texas, USA)	Avare Biotech Team (Dubai, UAE)
Owner, Timbercreek Veterinary Hospital (USA) "My real job for 40 years is Veterinary Medicine. I enjoy equine lameness and reproduction"	Veterinary Technician, Timbercreek Veterinary Hospital Pre-vet student, West Texas A&M University	Maksim Mershiev, The CEO&Co-Founder of Avare Biotech
Signature: Signed by: 4077863F342B4D7...	Signature: Signed by: Kylee Brown 7F41F20B5B08416...	Signature: Signed by: Maksim Mershiev 325996BF0B2B43A...

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Executive Summary / Conclusions

- The smartphone-based MAKSA technology proved to be simple and intuitive to use for both fresh and frozen-thawed equine semen samples, allowing rapid on-site evaluation.
- Motility results reflected the real biological situation and remained consistent with expectations from standard laboratory assessment.
- Concentration measurements were influenced by the use of regular lab slides, indicating the need for optimized slide geometry and volume control for more accurate quantification.
- Overall, the experience of working with the Avare Biotech app powered by MAKSA technology was excellent, demonstrating strong potential for routine field use and further technical refinement.

Objective

To validate the smartphone-based MAKSA (Mobile-Assisted Key Semen Analysis) technology powering Avare Biotech App for **fresh and frozen equine semen** evaluation against conventional CASA and Nucleocounter methods.

Materials and Methods

- Stallions were collected onsight, semen was extended with INRA 96 Equine Semen Extender and equal samples were taken for evaluation by the MAKSA and Hamilton Thorne IVOS II CASA machines.
- The MAKSA videos were captured with an IOS 18.6.2 device zoomed 3x and an Olympus CH-2 microscope with 200x magnification and phase contrast.
- Multiple videos were taken per collection using the MAKSA for increased accuracy.
- Results were collected and filed for comparison between the two machines.
- Regular lab slides and cover slips were used (NOT special CASA slides)

Table 1MAKSA results comparison

Tests done	Key Semen Parameters	MAKSA	CASA motility	Difference, %
Stallion 1	Concentration M/ml	22.4	33	47%
	Total Motility %	75	76	1%
	Progressive Motility %	53	53	0%
Stallion 2	Concentration M/ml	28.8	55.8	94%
	Total Motility %	96	92	-4%

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Stallion 3	Progressive Motility %	65	64	-2%
	Concentration M/ml	52.8	142	169%
	Total Motility %	89	77	-13%
	Progressive Motility %	64	50	-22%
Stallion 4	Concentration M/ml	67.2	287.5	328%
	Total Motility %	78	83	6%
	Progressive Motility %	56	60	7%
Stallion 5 (Bobby, fresh)	Concentration M/ml	41.6	159	282%
	Total Motility %	71	89	25%
	Progressive Motility %	52	59	13%
Stallion 5 (Bobby, frozen)	Concentration M/ml	65.6	46.2	-30%
	Total Motility %	53	62	17%
	Progressive Motility %	41	45	10%
Stallion 6	Concentration M/ml	25.6	82.5	222%
	Total Motility %	85	86	1%
	Progressive Motility %	55	61	11%

Discussion / Expert Insights

- Total and progressive motility correlates well with CASA readings
- Concentration values are always fewer and require adjustment via training the software with fresh-diluted equine semen data (videos).
- Overall, MAKSA proved stable, reliable, and user-friendly under real-world field conditions
- To improve users' journey it's recommended to make videos inside the app under the supervision of the app

Recommendations

- MAKSA results for frozen semen were different between each individual run and software may need further adjustments to be able to accurately analyze semen in egg based frozen semen extenders due to the higher protein concentration.

Appendix

• Screenshot of the semen videos made and CASA results
• MAKSA PDF report samples

Appendix – Equine semen video screenshots and the pdf report example of each sample tested

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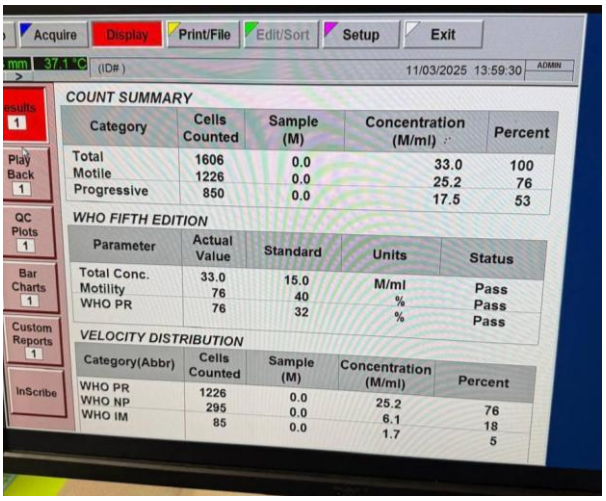
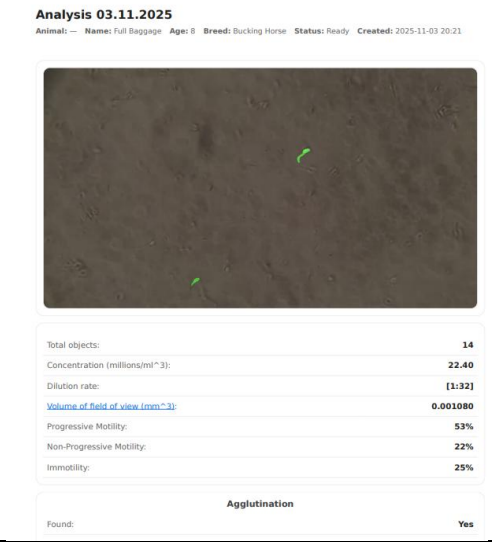
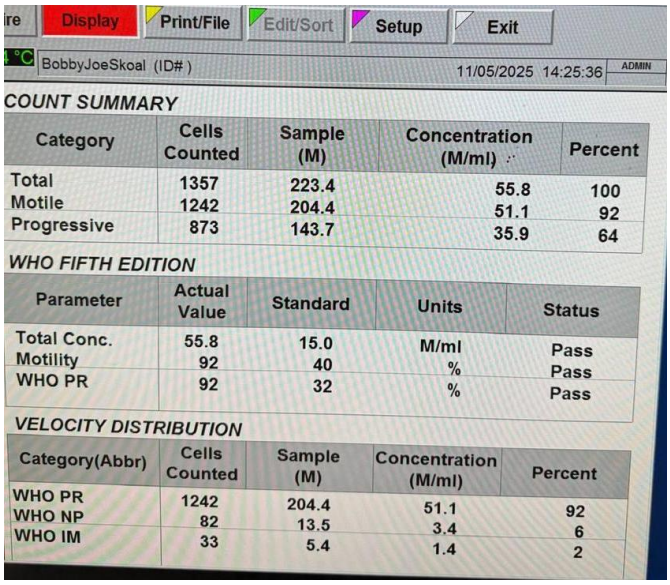
Stallion 1: CASA readings



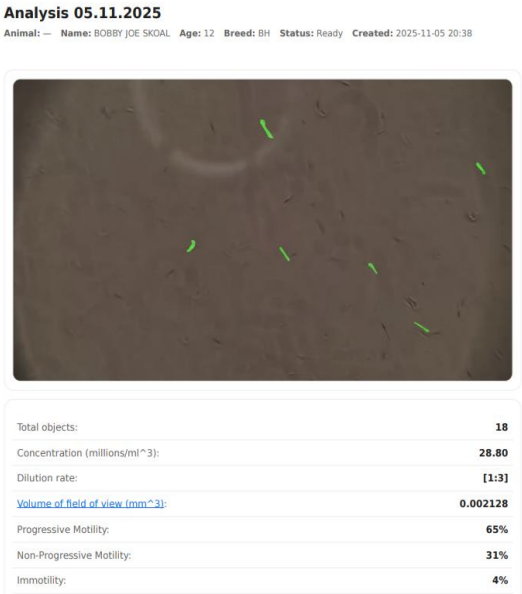
Stallion 1: MAKSA values



Stallion 2: CASA readings



Stallion 2: MAKSA values



Stallion 3: CASA readings

Stallion 3: MAKSA values

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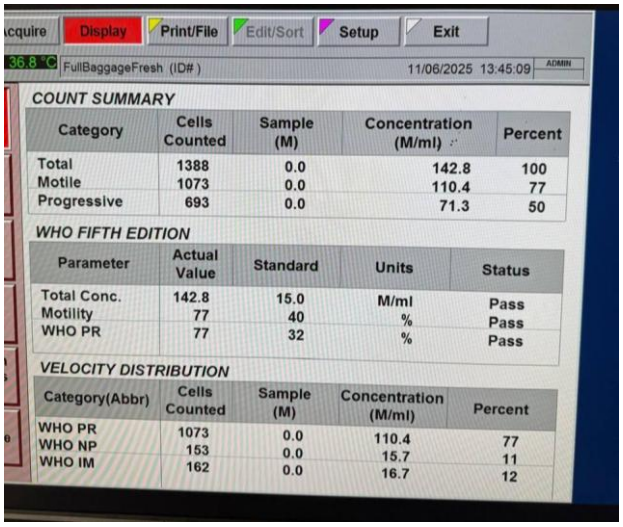
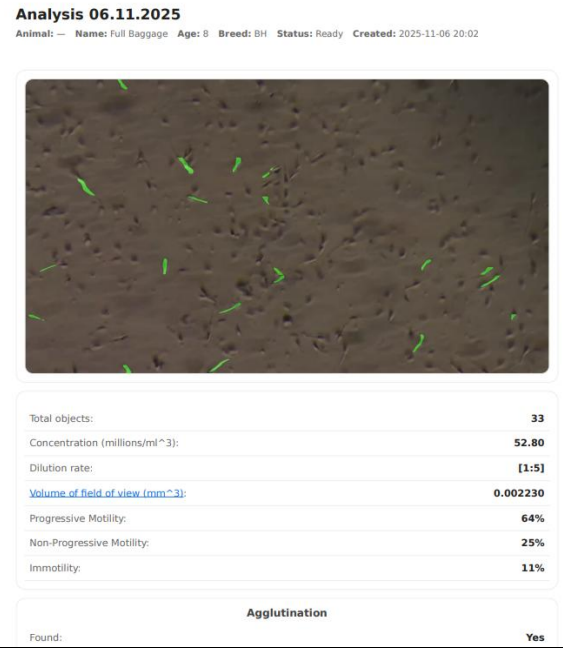
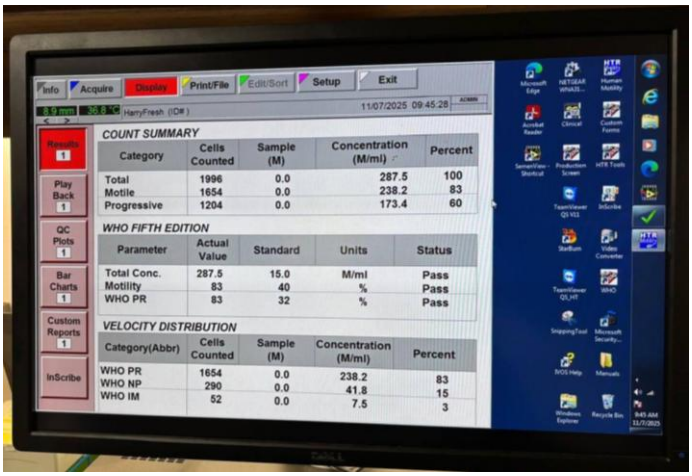


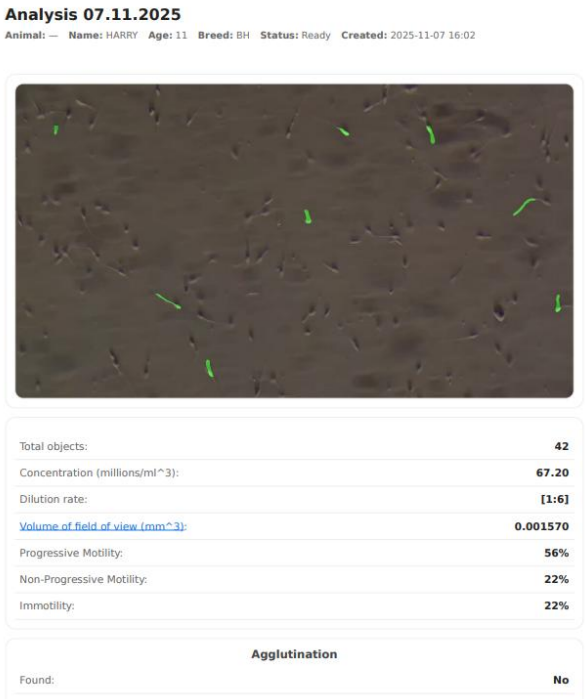
Stallion 4: CASA readings



Stallion 4: MAKSA values



Stallion 5 (fresh Bobby): CASA readings



Stallion 5(fresh Bobby): MAKSA values

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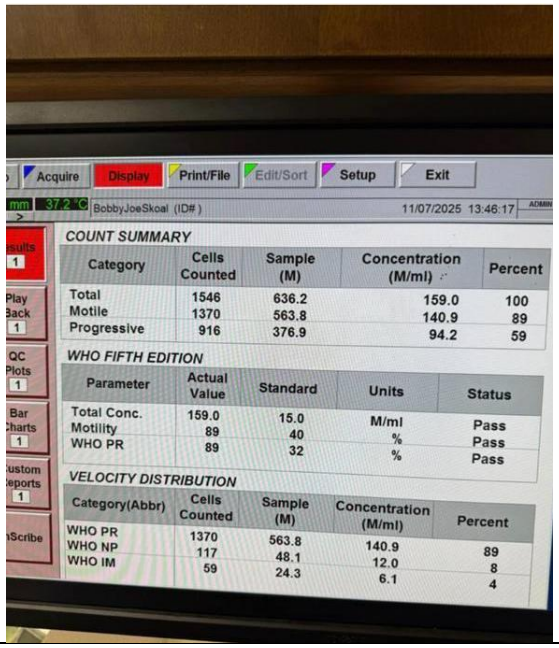
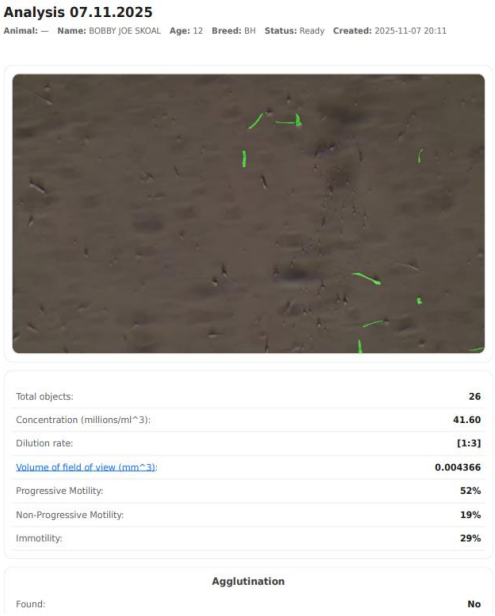
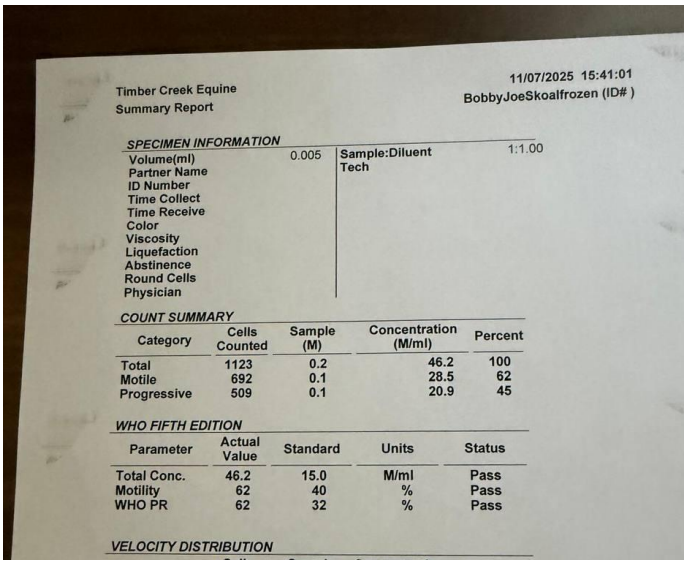


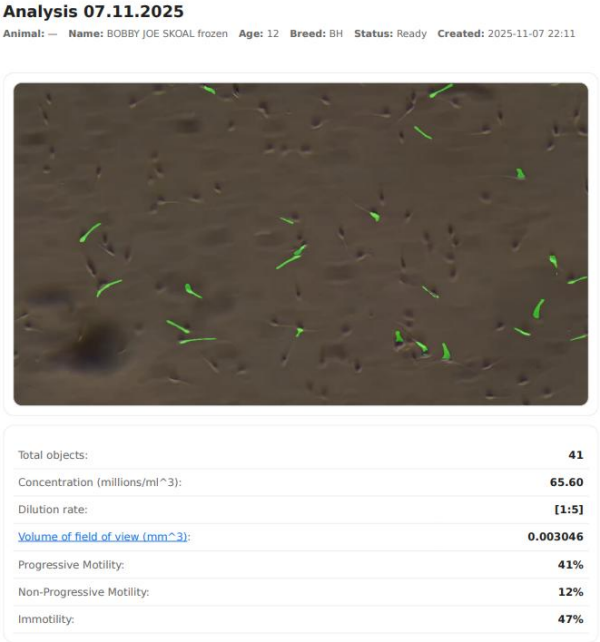
Stallion 5 (frozen Bobby): CASA readings



Stallion 5(frozen Bobby): MAKSA values



Stallion 6: CASA readings



Stallion 6: MAKSA values

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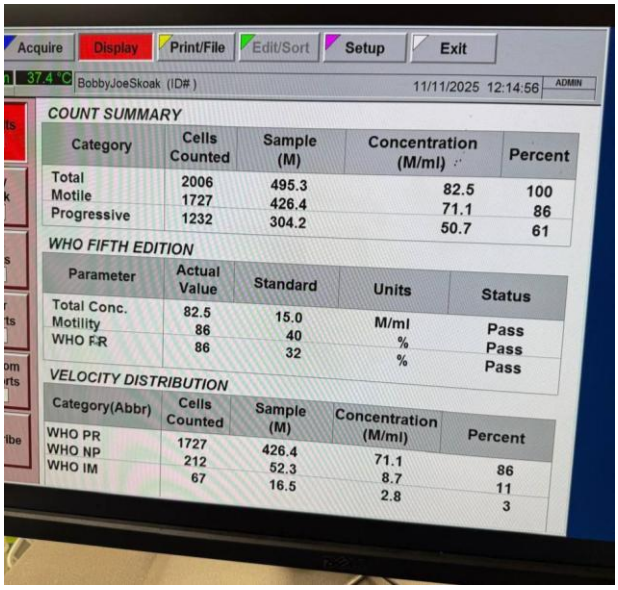


Acquire | Display | Print/File | Edit/Sort | Setup | Exit

37.4 °C BobbyJoeSkoak (ID#) 11/11/2025 12:14:56 ADMIN

COUNT SUMMARY

Category	Cells Counted	Sample (M)	Concentration (M/ml)	Percent
Total	2006	495.3	82.5	100
Motile	1727	426.4	71.1	86
Progressive	1232	304.2	50.7	61

WHO FIFTH EDITION

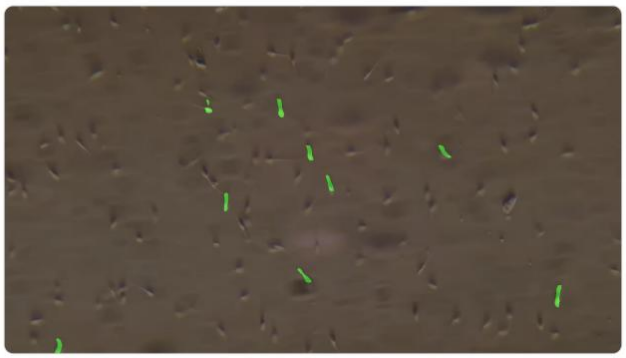
Parameter	Actual Value	Standard	Units	Status
Total Conc.	82.5	15.0	M/ml	Pass
Motility	86	40	%	Pass
WHO FR	86	32	%	Pass

VELOCITY DISTRIBUTION

Category(Abbr)	Cells Counted	Sample (M)	Concentration (M/ml)	Percent
WHO PR	1727	426.4	71.1	86
WHO NP	212	52.3	8.7	11
WHO IM	67	16.5	2.8	3

Analysis 11.11.2025

Animal: — **Name:** BOBBY JOE SKOAL **Age:** 10 **Breed:** BH **Status:** Ready **Created:** 2025-11-11 18:07



Total objects:	16
Concentration (millions/ml^3):	25.60
Dilution rate:	[1:3]
Volume of field of view (mm^3):	0.001676
Progressive Motility:	55%
Non-Progressive Motility:	30%
Immotility:	15%

Agglutination

Found:	No